

Depletion / recovery plug-in

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Status: draft

Purpose

The depletion / recovery plug-in collects Ecosim output from a range of EwE models to explore how the ecosystem recovers in the long term after time series perturbations.

User interface

The depletion / recovery plug-in can be activated through EwE6 > Menu > Tools > Depletion / Recovery..., which will bring up the following screen (Fig. 1):

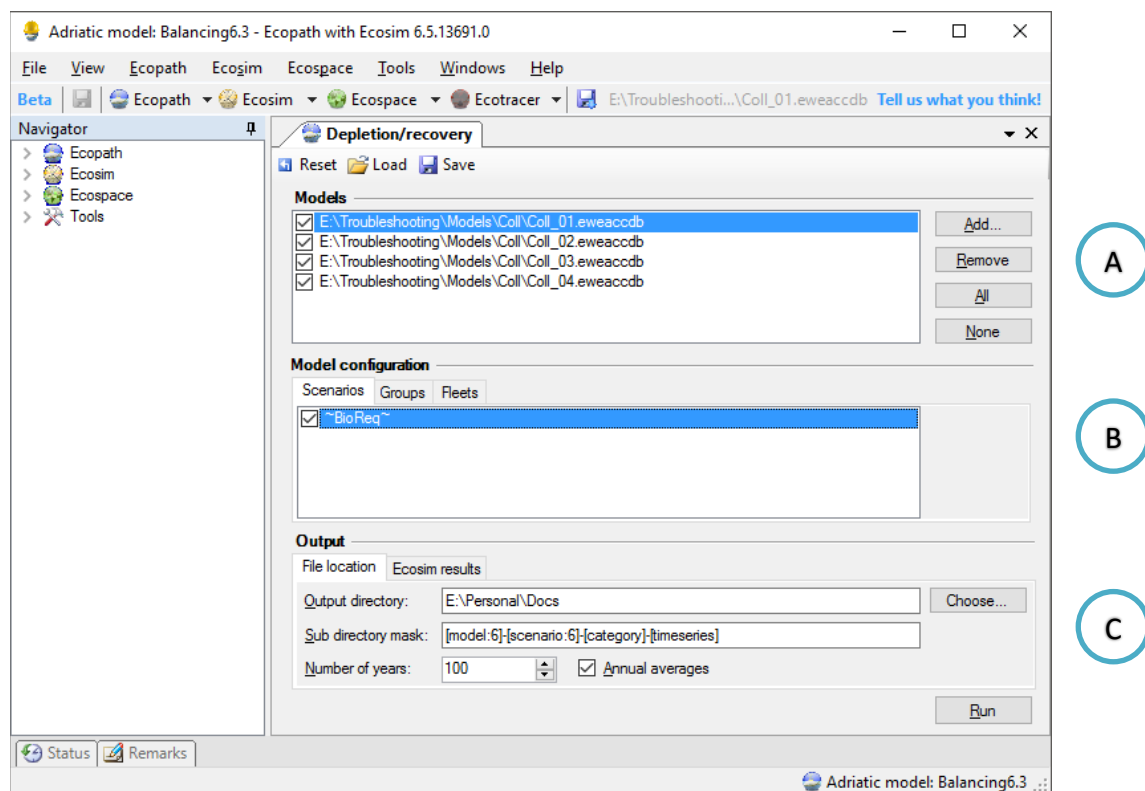


Figure 1 - The main interface of the Depletion / Recovery plug-in

This screen is organized as follows:

Main toolbar

The toolbar (top) offers the ability to save and load depletion / recovery run configurations. Here users can also reset the plug-in to a pristine layout, erasing all configuration information.

Section A, models

Here users manage which EwE models to run for the analysis. Models can be added by browsing or drag and drop, and models are deleted here. Note that only checked models will be included in the run. Selecting a model will load the model in the EwE user interface and will enable section B.

Section B, model configuration

In this section users manage how a model selected in section A should be executed by the depletion / recovery engine. More details about running will follow below.

First, in the scenario tab, users select which Ecosim scenarios must be executed from the selected model (Fig. 2). Note that only checked scenarios will be included in the run.

Second, in the groups tab, users can assign functional groups to any of the predetermined group categories. A functional group can be member of multiple categories (Fig. 3).

Last, in the fleets tab, users can assign fleets to predetermined fleet categories (Fig. 4).

Section C, output

Here users define which Ecosim model parameters are collected (Fig. 5) and where these results are written to (Fig. 6).

The File Location tab (Fig 6.) allows users to specify an output directory and a subdirectory mask where depletion / recovery results are written to. Underneath the indicated “Output directory” the plug-in will generate a subfolder structure which can be designed in the “sub directory mask” field. This field accepts a few keywords listed in table 1. If followed by a colon and a number, keywords will be truncated to indicated number. Please note that keywords do NOT contain spaces!

In this tab users can also specify a custom run length that overrides the settings specified in time series. The default depletion / recovery plug-in run length is set to 100 years.

Running the depletion / recovery engine

Schematically, the engine will run the model / scenario / category configuration as follows:

```
For every enabled model
  Load the model
  For every enabled scenario
    Load the scenario
    For every time series dataset
      Load the time series dataset
      For every category that has functional groups assigned to it
        Enable only time series that affect that category
        Run Ecosim
        Write Ecosim results
        Write network analysis results
      For every category that has fleets assigned to it
        Enable only time series that affect that category
        Run Ecosim
        Write Ecosim results
        Write network analysis results
```

Model configuration

Scenarios Groups Fleets

☒ ~BioReq~

Figure 2 – Selecting scenarios

Output

File location: Ecosim results

☒ Biomass ☒ Mortality ☒ Catch ☒ TL
☒ ConsumptionBiomass ☒ FeedingTime ☒ Value ☒ TLC
☒ PredationMortality ☒ Prey ☒ AvgWeightOrProdCor ☒ KemptonQ

Figure 5 - Configuring which outputs to collect

Model configuration

Scenarios Groups Fleets

category	# groups
DemPP	0
PelPP	0
Inv	10

☒ 10: Norway lobster
☒ 11: Mantis shrimp
☒ 12: Crabs
☒ 13: Benthic cephalop.
☒ 14: Benthop. cephalop.
☒ 15: Hake (1)

Figure 3 - Categorizing functional groups

Output

File location: Ecosim results

Output directory: E:\Personal\Docs

Sub directory mask: [model:6]-[scenario:6]-[category]-[timeseries]

Number of years: 100

Figure 6 – Configuring output settings

Model configuration

Scenarios Groups Fleets

category	# fleets
PurSein	0
Tuna	1
DriftNet	0

☒ 1: Strascico
☒ 2: Rapido
☒ 3: Volante
☒ 4: Lampara
☒ 5: Tuna fleets

Figure 4 - Categorizing fleets

Table 1 - Supported subdirectory fields

Keyword	Meaning	Examples
model	The name of the model as specified in the Ecopath Parameters form	{model} {model:10}
scenario	The name of the scenario as specified in the Ecosim Parameters form	{scenario}
category	The name of a group or fleet category as shown in the group and fleet tabs (section B, see Figs 3 and 4)	{category}
timeseries	The name of the time series dataset as specified in Ecosim > Time series	{timeseries} {timeseries:12}